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Equation of a Straight Line

Every line question needs a gradient and a point.
Then $y - y_1 = m(x - x_1)$ — and perpendicular means flip and change sign.

THE TOPIC ON ONE MAP

Equation of a Straight Line

01 — Gradient + point = line

$ax + by + c = 0$ through A and B

02 — Parallel and perpendicular

$m_1 m_2 = -1$ perpendicular bisector

03 — Intersections, areas, shapes

meet at P area of a triangle parallelogram

04 — Lines with curves and models

tangent parallel to l_1 $V = pt + q$

THE THREE TOOLS

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y - y_1 = m(x - x_1)$$

$$m_1 m_2 = -1$$

01 Gradient + Point = Line

Find the gradient, pick a point, write $y - y_1 = m(x - x_1)$, then the form they ask for.

WHEN YOU SEE → YOU DO

“passes through A and B”

→ $m = \frac{\Delta y}{\Delta x}$, then one of the points

“in the form $ax + by + c = 0$ where a, b, c are integers”

→ clear fractions, everything on one side

“find the gradient of $2x - 5y + 7 = 0$ ”

→ make y the subject: $m = \frac{2}{5}$

“ $H = at + b$ ”, two data points

→ gradient = change in H over change in t

WORKED MODEL • JANUARY 2026

$A(-1, 8), B(7, 2)$ on L_1 .

1 $m = \frac{2 - 8}{7 - (-1)} = -\frac{3}{4}$

gradient

2 $y - 8 = -\frac{3}{4}(x + 1)$

point A

ANSWER $3x + 4y - 29 = 0$

IDEA 01

Practise: Gradient + Point = Line

My questions first, then real past papers.

Q1 DR ESLAM Two points

P1 • TOPIC 8 3 marks

Find the equation of the line through (2, 1) and (4, 7) in the form $y = mx + c$.

PAST PAPERS

Q2 **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q2**P1 · TOPIC 8** **5 marks**

2. A tree was planted in the ground.

Exactly 2 years after it was planted, the height of the tree was 1.85 m.

Exactly 7 years after it was planted, the height of the tree was 3.45 m.

Given that the height, H metres, of the tree, t years after it was planted in the ground, can be modelled by the equation

$$H = at + b$$

where a and b are constants,

- (a) find the value of a and the value of b .

(4)

- (b) State, according to the model, the height of the tree when it was planted.

(1)

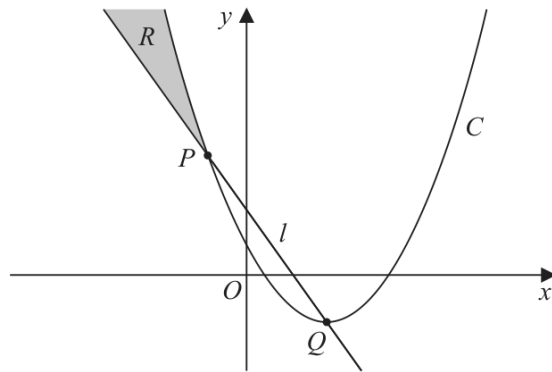
(Total 5 marks)

Q3**PAST PAPER**

Pearson IAL WMA11 · January 2021 · Q4

P1 · TOPIC 8 8 marks

4.

**Figure 2**

The points P and Q , as shown in Figure 2, have coordinates $(-2, 13)$ and $(4, -5)$ respectively.

The straight line l passes through P and Q .

- (a) Find an equation for l , writing your answer in the form $y = mx + c$, where m and c are integers to be found. (3)

The quadratic curve C passes through P and has a minimum point at Q .

- (b) Find an equation for C . (3)

The region R , shown shaded in Figure 2, lies in the second quadrant and is bounded by C and l only.

- (c) Use inequalities to define region R . (2)

(Total 8 marks)

Q4 **PAST PAPER** Pearson IAL WMA11 · October 2022 · Q3**P1 · TOPIC 8** **5 marks**

3. The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, £ V , of the company is modelled by the equation

$$V = pt + q$$

where t is the number of years after monitoring began and p and q are constants.

- (a) Find the value of p and the value of q .

(3)

Exactly T years after monitoring began, the share price was £2.50

- (b) Find the value of T , according to the model, giving your answer to one decimal place.

(2)

(Total for Question 3 is 5 marks)

Q5**PAST PAPER**

Pearson IAL WMA11 · January 2026 · Q4

P1 · TOPIC 8**9 marks**

4. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The point $A(-1, 8)$ and the point $B(7, 2)$ lie on the line L_1

- (a) Find an equation for L_1 in the form $ax + by + c = 0$, where a , b and c are integers. (3)

The line L_2

- passes through A and is perpendicular to L_1
- cuts the x -axis at the point C

- (b) Find the coordinates of C . (3)

- (c) Hence find the area of triangle ABC . (3)

(Total for Question 4 is 9 marks)

02 Parallel and Perpendicular

Parallel: same gradient. Perpendicular: flip and change the sign, $m_2 = -\frac{1}{m_1}$.

WHEN YOU SEE → YOU DO

“parallel to l_1 ”

→ same m

“perpendicular to l_1 ”

→ $m_2 = -\frac{1}{m_1}$

“prove angle $PQR = 90^\circ$ ”

→ $m_{PQ} \times m_{QR} = -1$

“perpendicular bisector”, “through the midpoint of AB ”

→ midpoint $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

“shortest distance from the line to the origin”

→ perpendicular from O , then $\sqrt{x^2 + y^2}$

WORKED MODEL • JUNE 2025

$l_1 : x - 2y + 25 = 0$; l_2 through O , perpendicular.

1 $m_1 = \frac{1}{2} \Rightarrow m_2 = -2: y = -2x$

perpendicular

2 $x - 2(-2x) + 25 = 0 \Rightarrow P(-5, 10)$

intersect

ANSWER distance $OP = \sqrt{25 + 100} = 5\sqrt{5}$

IDEA 02

Practise: Parallel and Perpendicular

My questions first, then real past papers.

Q6 DR ESLAM Perpendicular

P1 • TOPIC 8 3 marks

Find the line through (2, 1) perpendicular to $y = 3x - 5$, in the form $ax + by + c = 0$.

PAST PAPERS

Q7 PAST PAPER Pearson IAL WMA11 · January 2022 · Q8

P1 · TOPIC 8 9 marks

8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

- (a) Find the gradient of l_1 (1)

Given that

- the point A has coordinates $(6, -2)$
 - the line l_2 passes through A and is perpendicular to l_1
- (b) find the equation of l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found. (3)

The lines l_1 and l_2 intersect at the point M .

- (c) Using algebra and showing all your working, find the coordinates of M .

(Solutions relying on calculator technology are not acceptable.) (3)

Given that the diagonals of a square $ABCD$ meet at M ,

- (d) find the coordinates of the point C . (2)

(Total 9 marks)

Q8 **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q2**P1 · TOPIC 8** **5 marks**

2. The points P , Q and R have coordinates $(-3, 7)$, $(9, 11)$ and $(12, 2)$ respectively.

(a) Prove that angle $PQR = 90^\circ$

(3)

Given that the point S is such that $PQRS$ forms a rectangle,

(b) find the coordinates of S .

(2)

(Total for Question 2 is 5 marks)

Q9**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q1

P1 · TOPIC 8 6 marks

1. The line l_1 passes through the point $A(-5, 20)$ and the point $B(3, -4)$.

- (a) Find an equation for l_1 giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

The line l_2 is perpendicular to l_1 and passes through the midpoint of AB

- (b) Find an equation for l_2 giving your answer in the form $px + qy + r = 0$, where p , q and r are integers.

(3)**(Total for Question 1 is 6 marks)**

Q10**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q2

P1 · TOPIC 8 5 marks

2. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Given that

- the point A has coordinates $(-2\sqrt{3}, 5)$
 - the point B has coordinates $(7\sqrt{3}, 8)$
 - the straight line l_1 passes through A and B
- (a) show that the gradient of l_1 is $p\sqrt{3}$, where p is a rational constant to be found.
You must show each step of your working.

(2)

The straight line l_2 is perpendicular to l_1 and passes through A .

- (b) Find the equation of l_2 , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)**(Total for Question 2 is 5 marks)**

Q11**PAST PAPER**

Pearson IAL WMA11 • June 2025 • Q5

P1 • TOPIC 8 7 marks

5. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The line l_1 has equation

$$x - 2y + 25 = 0$$

The line l_2 passes through the origin and is perpendicular to l_1

- (a) Find an equation for l_2

(2)

The lines l_1 and l_2 intersect at the point P .

- (b) Use algebra to find the coordinates of P .

(3)

- (c) Hence find the shortest distance from l_1 to the origin.

Write your answer as a fully simplified surd.

(2)**(Total for Question 5 is 7 marks)**

03 Intersections, Areas, Shapes

Two lines meet: solve simultaneously. Areas: find the vertices, then base $\times$ height, or split into triangles.

WHEN YOU SEE $\rightarrow$ YOU DO

“the lines meet at P ”

$\rightarrow$ solve the two equations

“cuts the x -axis”

$\rightarrow y = 0$ · “the y -axis” $\rightarrow x = 0$

“area of triangle OPQ ”, “area of MPN ”

$\rightarrow$ sketch, base on an axis, height = the other coordinate

“ $PQRS$ is a parallelogram / rectangle / square”

$\rightarrow S = P + R - Q$ · diagonals meet at the midpoint

“both possible pairs for C ”, “length $BC = 5\sqrt{5}$ ”

$\rightarrow$ distance formula $\rightarrow$ quadratic $\rightarrow$ two answers

WORKED MODEL · JUNE 2026

$l_1 : y = 10 - 3x$; $l_2 \perp l_1$ meets it at $x = 4$.

1 $P(4, -2)$, $m_2 = \frac{1}{3}$: $x - 3y - 10 = 0$

perpendicular through P

2 $M(0, 10)$, $N(0, -\frac{10}{3})$

y -axis

ANSWER area = $\frac{1}{2} \cdot \frac{40}{3} \cdot 4 = \frac{80}{3}$

IDEA 03

Practise: Intersections, Areas, Shapes

My questions first, then real past papers.

Q12 **DR ESLAM** Where they meet

P1 · TOPIC 8 2 marks

Find where $y = 2x + 1$ meets $x + y = 7$.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA11 · January 2020 · Q6

P1 · TOPIC 8 8 marks

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

- (a) Find the equation of l_2 , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

The line l_1 cuts the x -axis at P and the y -axis at Q .

Given that $PQRS$ is a parallelogram, find

- (b) the area of $PQRS$,

(3)

- (c) the coordinates of S .

(2)

(Total 8 marks)

Q14 PAST PAPER Pearson IAL WMA11 · October 2021 · Q5

P1 · TOPIC 8 8 marks

5. The line l_1 has equation $3y - 2x = 30$

The line l_2 passes through the point $A(24, 0)$ and is perpendicular to l_1

Lines l_1 and l_2 meet at the point P

- (a) Find, using algebra and showing your working, the coordinates of P . (5)

Given that l_1 meets the x -axis at the point B ,

- (b) find the area of triangle BPA . (3)

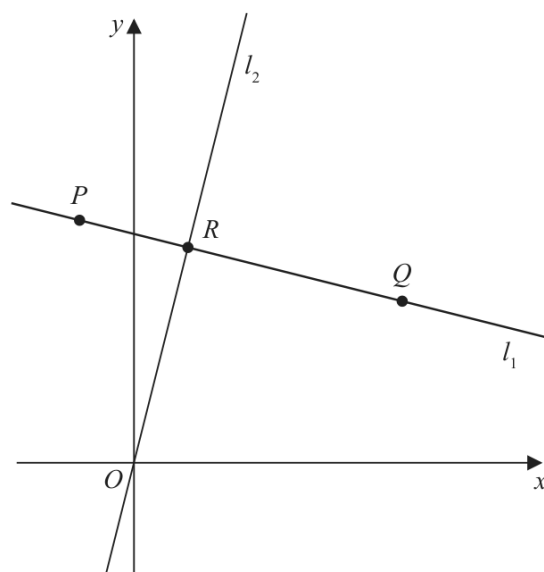
(Total 8 marks)

Q15**PAST PAPER**

Pearson IAL WMA11 · January 2024 · Q5

P1 · TOPIC 8 10 marks

5.

**Figure 1****In this question you must show all stages of your working.****Solutions relying on calculator technology are not acceptable.**

The straight line l_1 , shown in Figure 1, passes through the points $P(-2, 9)$ and $Q(10, 6)$.

- (a) Find the equation of l_1 , giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The straight line l_2 passes through the origin O and is perpendicular to l_1 .

The lines l_1 and l_2 meet at the point R as shown in Figure 1.

- (b) Find the coordinates of R

(4)

- (c) Find the exact area of triangle OPQ .

(3)**(Total for Question 5 is 10 marks)**

Q16 PAST PAPER Pearson IAL WMA11 · June 2024 · Q9

P1 · TOPIC 8 9 marks

9. Given that

- the point A has coordinates $(4, 2)$
 - the point B has coordinates $(15, 7)$
 - the line l_1 passes through A and B
- (a) find an equation for l_1 , giving your answer in the form $px + qy + r = 0$ where p , q and r are integers to be found.

(3)

The line l_2 passes through A and is parallel to the x -axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

- (b) Find both possible pairs of coordinates of the point C .

(4)

- (c) Hence find the minimum possible area of triangle ABC .

(2)

(Total for Question 9 is 9 marks)

Q17 PAST PAPER Pearson IAL WMA11 · June 2026 · Q2

P1 · TOPIC 8 6 marks

2. The line l_1 has equation $y = 10 - 3x$

The line l_2

- is perpendicular to l_1
- intersects l_1 at the point P with x coordinate 4

(a) Find an equation for l_2 writing the answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

Given that

- line l_1 cuts the y -axis at the point M
- line l_2 cuts the y -axis at the point N

(b) find the area of triangle MPN , making your method clear.

(3)

Question 2 continued

(Total for Question 2 is 6 marks)

04 Lines with Curves and Models

Lines join quadratics: the turning point gives the curve, the line through two points gives the region, inequalities define it.

WHEN YOU SEE → YOU DO

“maximum point $(-2, 12)$, cuts at -5 ”

→ $f(x) = 12 - k(x + 2)^2$, the root gives k

“a line touches C and is parallel to l_1 ”

→ $\frac{dy}{dx}$ = that gradient, or discriminant = 0

“use inequalities to define R ”

→ one inequality per edge

“ $V = pt + q$... find T ”

→ two data points → p, q → substitute

WORKED MODEL • JUNE 2022

Max $(-2, 12)$, cuts at $x = -5$.

1 $f(x) = 12 - k(x + 2)^2$

turning point form

2 $0 = 12 - 9k \Rightarrow k = \frac{4}{3}$

root -5

ANSWER $f(x) = 12 - \frac{4}{3}(x + 2)^2$

IDEA 04

Practise: Lines with Curves and Models

My questions first, then real past papers.

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA11 · October 2021 · Q3

P1 · TOPIC 8 6 marks

3. (i) Solve

$$\frac{3}{x} > 4$$

(3)

(ii)

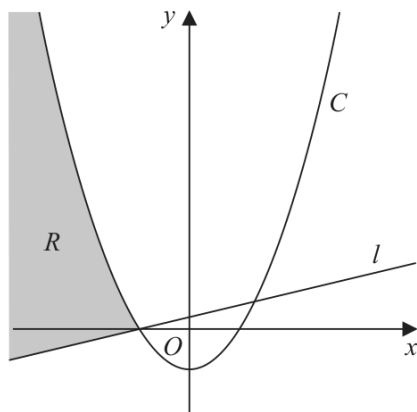


Figure 1

Figure 1 shows a sketch of the curve C and the straight line l .

The infinite region R , shown shaded in Figure 1, lies in quadrants 2 and 3 and is bounded by C and l only.

Given that

- l has a gradient of 3
- C has equation $y = 2x^2 - 50$
- C and l intersect on the negative x -axis

use inequalities to define the region R .

(3)

(Total 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · October 2021 · Q8

P1 · TOPIC 8 10 marks

8.

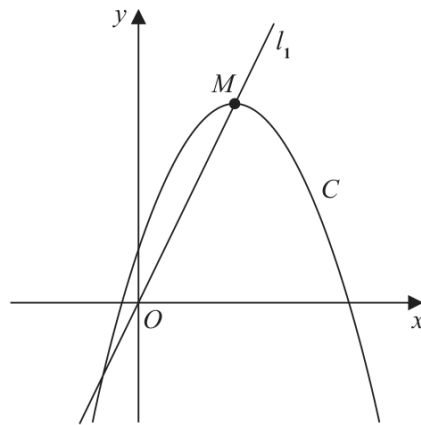
**Figure 4**

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C .

- (a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a , b and c are constants to be found.

- (ii) Hence, or otherwise, state the coordinates of M .

(5)

The line l_1 passes through O and M , as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

- (b) Find an equation for l_2

(5)**(Total 10 marks)**

Q20 PAST PAPER Pearson IAL WMA11 · June 2022 · Q5

P1 · TOPIC 8 9 marks

5. The curve C has equation $y = f(x)$

Given that

- $f(x)$ is a quadratic expression
- the maximum turning point on C has coordinates $(-2, 12)$
- C cuts the negative x -axis at -5

- (a) find $f(x)$

(4)

The line l_1 has equation $y = \frac{4}{5}x$

Given that the line l_2 is perpendicular to l_1 and passes through $(-5, 0)$

- (b) find an equation for l_2 , writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

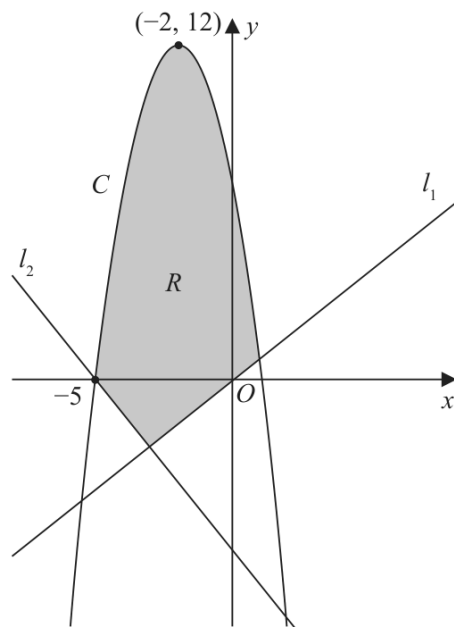


Figure 2

Figure 2 shows a sketch of the curve C and the lines l_1 and l_2

- (c) Define the region R , shown shaded in Figure 2, using inequalities.

(2)

(Total 9 marks)

10.

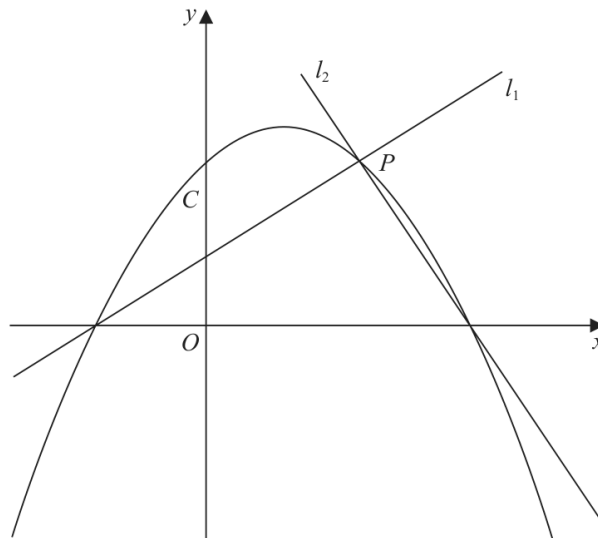


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \quad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x -axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b , the coordinates of P

(3)

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x -axis,

(b) find, in terms of b , an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

(d) Hence, or otherwise, find the value of b

(Total for Question 10 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 The line** $m = \frac{\Delta y}{\Delta x}$, $y - y_1 = m(x - x_1)$, then the asked form.
- 02 $\parallel$ and $\perp$** Same $m \cdot m_2 = -\frac{1}{m_1}$. Midpoint for bisectors.
- 03 Intersections & areas** Solve together. Sketch, base on an axis. $S = P + R - Q$.
- 04 With curves** Turning point form + a root. Regions: one inequality per edge.

BOTTOM LINE Gradient + a point → the line. Perpendicular → flip and change the sign.

Answers

Equation of a Straight Line • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • GRADIENT + POINT = LINE

- Q1** Dr Eslam • Two points $m = 3: y = 3x - 5$
- Q2** Pearson IAL WMA11 • October 2019 • Q2 (a) $a = 0.32, b = 1.21$ (b) 1.21 m
- Q3** Pearson IAL WMA11 • January 2021 • Q4 (a) $y = 7 - 3x$ (b) $y = \frac{1}{2}(x - 4)^2 - 5$ (c) $y \geq 7 - 3x, y \leq \frac{1}{2}(x - 4)^2 - 5, x \leq 0$
- Q4** Pearson IAL WMA11 • October 2022 • Q3 (a) $p = 0.3, q = 0.15$ (b) $T = 7.8$
- Q5** Pearson IAL WMA11 • January 2026 • Q4 (a) $3x + 4y - 29 = 0$ (b) $C(-7, 0)$ (c) 50

IDEA 02 • PARALLEL AND PERPENDICULAR

- Q6** Dr Eslam • Perpendicular $m = -\frac{1}{3}: x + 3y - 5 = 0$
- Q7** Pearson IAL WMA11 • January 2022 • Q8 (a) $\frac{2}{5}$ (b) $y = -\frac{5}{2}x + 13$ (c) $M(4, 3)$ (d) $C(2, 8)$
- Q8** Pearson IAL WMA11 • January 2023 • Q2 (a) $m_{PQ} = \frac{1}{3}, m_{QR} = -3$, product -1 (b) $S(0, -2)$
- Q9** Pearson IAL WMA11 • October 2024 • Q1 (a) $y = -\frac{24}{5}x - 4$ (b) $5x - 24y + 187 = 0$
- Q10** Pearson IAL WMA11 • January 2025 • Q2 (a) $p = \frac{1}{9}$ (b) $y = -3\sqrt{3}x - 13$
- Q11** Pearson IAL WMA11 • June 2025 • Q5 (a) $y = -2x$ (b) $P(-5, 10)$ (c) $5\sqrt{5}$

IDEA 03 · INTERSECTIONS, AREAS, SHAPES

- Q12** Dr Eslam · Where they meet (2, 5)
- Q13** Pearson IAL WMA11 · January 2020 · Q6 (a) $3x - 4y - 24 = 0$ (b) $\frac{220}{3}$ (c) $S\left(\frac{4}{3}, -5\right)$
- Q14** Pearson IAL WMA11 · October 2021 · Q5 (a) $P(12, 18)$ (b) 351
- Q15** Pearson IAL WMA11 · January 2024 · Q5 (a) $y = -\frac{1}{4}x + \frac{17}{2}$ (b) $R(2, 8)$ (c) 51
- Q16** Pearson IAL WMA11 · June 2024 · Q9 (a) $5x - 11y + 2 = 0$ (b) $(25, 2), (5, 2)$ (c) $\frac{5}{2}$
- Q17** Pearson IAL WMA11 · June 2026 · Q2 (a) $x - 3y - 10 = 0$ (b) $\frac{80}{3}$

IDEA 04 · LINES WITH CURVES AND MODELS

- Q18** Pearson IAL WMA11 · October 2021 · Q3 (i) $0 < x < \frac{3}{4}$ (ii) $3x + 15 \leq y \leq 2x^2 - 50, x \leq -5$
- Q19** Pearson IAL WMA11 · October 2021 · Q8 (a) $16 - 3(x - 2)^2; M(2, 16)$ (b) $y = 8x + \frac{16}{3}$
- Q20** Pearson IAL WMA11 · June 2022 · Q5 (a) $f(x) = 12 - \frac{4}{3}(x + 2)^2$ (b) $y = -\frac{5}{4}x - \frac{25}{4}$
- Q21** Pearson IAL WMA11 · June 2023 · Q10 (a)(i) $y = \frac{1}{2}x + 1$ (ii) $P\left(b - 2, \frac{b}{2}\right)$ (b) $y = -2x + 2b$
(d) $b = 8$